

MA1521 L1 Week 04 Mon 20260831

Sketch five level curves for each of the following functions.

Also, sketch the portion of the graph of the function lying in the first octant

a) $1 - x - y$ b) $\sqrt{x^2 + y^2}$ c) $x^2 + y^2$ d) $1 - x^2 - y^2$

Calculate the first partial derivatives of each of the following functions:

$$x^3y - 3xy^2 + 2y^2$$

$$\sin(3x + 2y)$$

$$x \ln(2x + y)$$

$$x^2z - 2yz^3$$

Verify that $f_{xy} = f_{yx}$ for each of the following:

- a) $x^m y^n$, (m, n positive integers) b) $\frac{x}{x+y}$ c) $\cos(x^2 + y)$

By using $f_{xy} = f_{yx}$, tell for what value of the constant a there exists a function $f(x, y)$ for which $f_x = axy + 3y^2$, $f_y = x^2 + 6xy$, and then using this value, find such a function by inspection.

Show the following functions $w = f(x, y)$ satisfy the equation $w_{xx} + w_{yy} = 0$ (called the two-dimensional *Laplace equation*):

a) $w = e^{ax} \sin ay$ (a constant)

b) $w = \ln(x^2 + y^2)$

Give the equation of the tangent plane to each of these surfaces at the point indicated.

a) $z = xy^2$, $(1, 1, 1)$ b) $w = y^2/x$, $(1, 2, 4)$

- a) Find the equation of the tangent plane to the cone $z = \sqrt{x^2 + y^2}$ at the point $P_0 : (x_0, y_0, z_0)$ on the cone.
- b) Write parametric equations for the ray from the origin passing through P_0 , and using them, show the ray lies on both the cone and the tangent plane at P_0 .